

Q1: For the following three-objective optimization problem, it is desired to determine the ideal and nadir solutions.

$$\begin{aligned} \text{Minimize } f_1(x) &= (1 + g(x_3)) \cos(\theta_1 \pi/2) \cos(\theta_2 \pi/2), \\ \text{Minimize } f_2(x) &= (1 + g(x_3)) \cos(\theta_1 \pi/2) \sin(\theta_2 \pi/2), \\ \text{Minimize } f_3(x) &= (1 + g(x_3)) \sin(\theta_1 \pi/2), \end{aligned}$$

$$\begin{aligned} \text{where } \theta_i &= \frac{x_i}{4(1 + g(x_3))} \quad \text{for } i = 1, 2 \\ g(x_3) &= (x_3 - 0.5)^2, \\ 0 \leq x_i &\leq 1, \quad \text{for } i = 1, 2, 3. \end{aligned}$$

Solution

For ideal solution we use:

$$f_m^* = \min_{X \in S} f_m(X), \quad m = 1, 2, 3 \implies \mathbf{z}^* = (f_1^*, f_2^*, f_3^*)^t \quad (1)$$

And the nadir is computed via the Payoff Table method ($\mathbf{x}^{*(m)}$ = minimizer of f_m):

$$z_m^{nad} = \max_{k \neq m} f_m(\mathbf{x}^{*(k)}), \quad m = 1, 2, 3 \quad (2)$$

Let rewrite objectives:

$$\text{[Redacted Objective 1]} \quad (3)$$

$$\text{[Redacted Objective 2]} \quad (4)$$

$$\text{[Redacted Objective 3]} \quad (5)$$

$$\text{[Redacted Objective 4]} \quad (6)$$

From $x_3 \in [0, 1]$:

$$\text{[Redacted Equation 7]} \quad (7)$$

R is minimized and maximized at:

$$\text{[Redacted Equation 8]} \quad (8)$$

The angular arguments lie in:

$$\text{[Redacted]} \quad (9)$$

so $\cos() > 0$, $\sin() \geq 0$, and all $f_m \geq 0$.

For find ideal solution, we do following steps.

Step 1: Minimize f_1 (find $\mathbf{x}^{*(1)}$)

From (4):

$$\text{[Redacted]} \quad (10)$$

All three factors R , $\cos\left(\frac{x_1\pi}{8R}\right)$, $\cos\left(\frac{x_2\pi}{8R}\right)$ are positive. To minimize f_1 :

(a) Minimize R :

$$\text{[Redacted]} \quad (11)$$

More directly: f_1 is increasing in R , so set R to its minimum:

$$\text{[Redacted]} \quad (12)$$

With $R = 1$, the angular arguments simplify to $\frac{x_i\pi}{8}$.

(b) Minimize $\cos\left(\frac{x_1\pi}{8}\right)$:

$\cos$ is decreasing on $\left[0, \frac{\pi}{8}\right]$, so maximize the argument:

$$\text{[Redacted]} \quad (13)$$

(c) Minimize $\cos\left(\frac{x_2\pi}{8}\right)$:

Same reasoning:

$$\text{[Redacted]} \quad (14)$$

Therefore $\mathbf{x}^{*(1)} = (1, 1, 0.5)$ and:

$$\text{[Redacted]} \quad (15)$$

$$\text{Using } \cos^2\left(\frac{\pi}{8}\right) = \frac{1 + \cos\left(\frac{\pi}{4}\right)}{2} = \frac{1 + \frac{\sqrt{2}}{2}}{2}:$$

$$\boxed{f_1^* = 0.8536} \quad (16)$$

Step 2: Minimize f_2 (find $\mathbf{x}^{*(2)}$)

From (5):

$$\sin \theta_1 \geq 0 \quad (17)$$

$\sin$ is increasing on $\left[0, \frac{\pi}{8}\right]$, so minimize the argument by setting:

$$\theta_1 = 0 \quad (18)$$

This drives $f_2 = 0$ regardless of x_1, x_3 . Since $f_2 \geq 0$ always, $f_2 = 0$ is indeed the global minimum.

Choose $\mathbf{x}^{*(2)} = (0, 0, 0.5)$:

$$f_2^* = 0 \quad (19)$$

Step 3: Minimize f_3 (find $\mathbf{x}^{*(3)}$)

From (6):

$$\sin \theta_2 \geq 0 \quad (20)$$

$\sin$ is increasing on $\left[0, \frac{\pi}{8}\right]$, so minimize the argument by setting:

$$\theta_2 = 0 \quad (21)$$

This drives $f_3 = 0$ regardless of x_2, x_3 . Since $f_3 \geq 0$ always, this is the global minimum.

Choose $\mathbf{x}^{*(3)} = (0, 0, 0.5)$:

$$f_3^* = 0 \quad (22)$$

Ideal Solution

$$\mathbf{z}^* = (0.8536, 0, 0)^t \quad (23)$$

For nadir solution, first evaluate all objectives at each minimizer. With $R = 1$ and $\mathbf{x}^{*(1)} = (1, 1, 0.5)$:

$$f_1(\mathbf{x}^{*(1)}) = 1 \quad (24)$$

$$f_2(\mathbf{x}^{*(1)}) = 1 \quad (25)$$

$$f_3(\mathbf{x}^{*(1)}) = 0.5 \quad (26)$$

With $\mathbf{x}^{*(2)} = (0, 0, 0.5)$, all angles = 0:

$$f_1(\mathbf{x}^{*(2)}) = 0 \quad (27)$$

With $\mathbf{x}^{*(3)} = (0, 0, 0.5)$, same as above:

$$\text{[Redacted]} \quad (28)$$

Payoff table:

$$\text{[Redacted Payoff Table]}$$

$$\text{[Redacted]} \quad (29)$$

$$\text{[Redacted]} \quad (30)$$

$$\text{[Redacted]} \quad (31)$$

Nadir Solution

$$\mathbf{z}^{nad} = (1, 0.3536, 0.3827)^t \quad (32)$$

Q2: Suppose that during the solution of the following multi-objective optimization problem:

$$\begin{aligned} &\text{Minimize } f_1 \\ &\text{Minimize } f_2 \\ &\text{Minimize } f_3 \end{aligned}$$

the following solutions have been obtained:

Soln. Id.	f_1	f_2	f_3
1	2	3	1
2	5	1	10
3	3	4	10
4	2	2	2
5	3	3	2
6	4	4	5

It is desired to sort the resulting solutions using the non-dominated sorting algorithm $O(MN^2)$

Solution

The population is $P = \{1, 2, 3, 4, 5, 6\}$ with $M = 3$ objectives and $N = 6$ solutions. Solution i dominates solution j if and only if:

$$f_k(i) \leq f_k(j) \quad \forall k \in \{1, 2, 3\}, \quad \text{and} \quad f_k(i) < f_k(j) \quad \text{for at least one } k \quad (33)$$

For each $i \in P$:

- n_i = number of solutions that dominate i , initialized to $n_i = 0$.
- S_i = set of solutions dominated by i , initialized to $S_i = \emptyset$.

We check all ordered pairs (i, j) , $i \neq j$, using (33).

Solution 1 vs. others: $\mathbf{f}(1) = (2, 3, 1)$

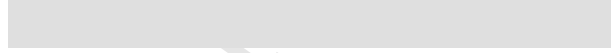
- 1 vs 2 $\rightarrow (2, 3, 1)$ vs $(5, 1, 10)$:



- 1 vs 3 $\rightarrow (2, 3, 1)$ vs $(3, 4, 10)$:



- 1 vs 4 $\rightarrow (2, 3, 1)$ vs $(2, 2, 2)$:



- 1 vs 5 $\rightarrow (2, 3, 1)$ vs $(3, 3, 2)$:



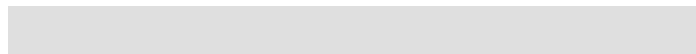
- 1 vs 6 $\rightarrow (2, 3, 1)$ vs $(4, 4, 5)$:



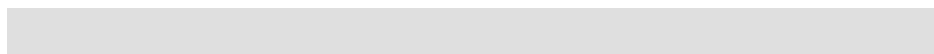
$$n_1 = 0, \quad S_1 = \{3, 5, 6\}$$

Solution 2 vs. others: $\mathbf{f}(2) = (5, 1, 10)$

- 2 vs 1 (checked above, neither)
- 2 vs 3 $\rightarrow (5, 1, 10)$ vs $(3, 4, 10)$:



- 2 vs 4 $\rightarrow (5, 1, 10)$ vs $(2, 2, 2)$:



- 2 vs 5 $\rightarrow (5, 1, 10)$ vs $(3, 3, 2)$:

- 2 vs 6 $\rightarrow (5, 1, 10)$ vs $(4, 4, 5)$:

$$n_2 = 0, \quad S_2 = \emptyset$$

Solution 3 vs. others: $\mathbf{f}(3) = (3, 4, 10)$

- 3 vs 1 : $1 \preceq 3$ (shown above) $\Rightarrow n_3 += 1$
- 3 vs 2 (checked above, neither)
- 3 vs 4 $\rightarrow (3, 4, 10)$ vs $(2, 2, 2)$:

- 3 vs 5 $\rightarrow (3, 4, 10)$ vs $(3, 3, 2)$:

- 3 vs 6 $\rightarrow (3, 4, 10)$ vs $(4, 4, 5)$:

$$n_3 = 3, \quad S_3 = \emptyset$$

Solution 4 vs. others: $\mathbf{f}(4) = (2, 2, 2)$

- 4 vs 1 $\rightarrow (2, 2, 2)$ vs $(2, 3, 1)$:

- 4 vs 2 $\rightarrow (2, 2, 2)$ vs $(5, 1, 10)$:

- 4 vs 3 $\rightarrow 4 \preceq 3$ (shown above) $\Rightarrow S_4 = S_4 \cup \{3\}$

- 4 vs 5 $\rightarrow (2, 2, 2)$ vs $(3, 3, 2)$:

- 4 vs 6 $\rightarrow (2, 2, 2)$ vs $(4, 4, 5)$:

$$n_4 = 0, \quad S_4 = \{3, 5, 6\}$$

Solution 5 vs. others: $\mathbf{f}(5) = (3, 3, 2)$

- 5 vs 1 $\rightarrow 1 \preceq 5$ (shown above) $\Rightarrow n_5 += 1$
- 5 vs 2 : (checked above, neither)
- 5 vs 3 $\rightarrow 5 \preceq 3$ (shown above) $\Rightarrow S_5 = S_5 \cup \{3\}$
- 5 vs 4 $\rightarrow 4 \preceq 5$ (shown above) $\Rightarrow n_5 += 1$
- 5 vs 6 $\rightarrow (3, 3, 2)$ vs $(4, 4, 5)$:

$$n_5 = 2, \quad S_5 = \{3, 6\}$$

Solution 6 vs. others: $\mathbf{f}(6) = (4, 4, 5)$

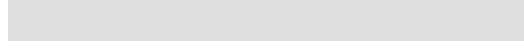
- 6 vs 1 $\rightarrow 1 \preceq 6$ (shown above) $\Rightarrow n_6 += 1$
- 6 vs 2 (checked above, neither)
- 6 vs 3 (checked above, neither)
- 6 vs 4 $\rightarrow 4 \preceq 6$ (shown above) $\Rightarrow n_6 += 1$
- 6 vs 5 $\rightarrow 5 \preceq 6$ (shown above) $\Rightarrow n_6 += 1$

$$n_6 = 3, \quad S_6 = \emptyset$$

First Front (F_1): Solutions with $n_i = 0$:

(34)

Second Front (F_2): For each $i \in F_1$ and each $j \in S_i$, decrement $n_j \leftarrow n_j - 1$:
 From $i = 1$, $S_1 = \{3, 5, 6\}$:



From $i = 2$, $S_2 = \emptyset$: no change.

From $i = 4$, $S_4 = \{3, 5, 6\}$:



Updated domination counts:



Solutions with $n_j = 0$ form the next front:



(35)

Third Front (F_3): For $i = 5$, $S_5 = \{3, 6\}$:



Solutions with $n_j = 0$:



(36)

Final Result

$$F_1 = \{1, 2, 4\}, \quad F_2 = \{5\}, \quad F_3 = \{3, 6\}$$

The Pareto front (first non-dominated front) is (34).

Q3: The following bi-objective optimization problem is given:

$$\begin{aligned} &\max \quad f_1(x_1, x_2) = x_1 + x_2 \\ &\max \quad f_2(x_1, x_2) = (x_1 + 2)^2 + (x_2)^2 \\ &\text{subject to} \\ &\quad 0 \leq x_1 \leq 1 \\ &\quad 0 \leq x_2 \leq 2 \end{aligned}$$

a) Using the weight vector $W = (w_1, 1 - w_1)^T$, it is desired to find the Pareto optimal solutions in terms of w_1 .

b) What is the Pareto optimal solution corresponding to $w = 0.5$?

Solution**Part (a):** Weighted Sum Scalarization

The weighted sum composite objective:

$$\text{[Redacted Equation]} \quad (37)$$

with $w_1 \in [0, 1]$.

$$\text{[Redacted Equation]}$$

$$\text{[Redacted Equation]}$$

Solving for the interior stationary point:

$$\text{[Redacted Equation]} \quad (38)$$

Since $w_1 \geq 0$ and $1 - w_1 > 0$, both expressions in (38) are ≤ -2 and < 0 respectively, which violate the bounds $x_1 \geq 0$, $x_2 \geq 0$. Therefore no interior maximum exists; the maximum of F lies on the boundary of the feasible region.

On the feasible box $[0, 1] \times [0, 2]$:

$$\text{[Redacted Equation]}$$

F is convex in (x_1, x_2) for $w_1 < 1$, so its maximum over the box is attained at a corner. The four corners of the feasible domain are:

$$\text{[Redacted Equation]} \quad (39)$$

$$\text{[Redacted Equation]} \quad (40)$$

$$\text{[Redacted Equation]} \quad (41)$$

$$\text{[Redacted Equation]} \quad (42)$$

Compare (42) with (40):

$$\text{[Redacted Equation]}$$

$$\text{[Redacted Equation]}$$

Compare (42) with (41):

$$\text{[Redacted Equation]}$$

$$\text{[Redacted Equation]}$$

Compare (42) with (39):

$$\text{[Redacted Equation]}$$

Therefore, for all $w_1 \in [0, 1]$, the corner $(x_1, x_2) = (1, 2)$ maximises F .

Pareto Optimal Solution

$$x_1^* = 1, \quad x_2^* = 2 \quad \text{for all } w_1 \in [0, 1] \quad (43)$$

with the corresponding objective values:

$$\quad \quad \quad (44)$$

Part (b): Solution for $w_1 = 0.5$

Setting $w_1 = 0.5$, the composite objective (37) becomes:

$$\quad \quad \quad (45)$$

From the result of Part (a), the solution (43) is independent of w_1 . Evaluating (45) at all corners to confirm:

$$\begin{aligned} & \quad \quad \quad \\ & \quad \quad \quad \\ & \quad \quad \quad \\ & \quad \quad \quad \end{aligned}$$

The maximum is attained at $(1, 2)$ as confirmed by (43).

$$w_1 = 0.5 : \quad x_1^* = 1, \quad x_2^* = 2, \quad f_1^* = 3, \quad f_2^* = 13 \quad (46)$$

Q4: It is desired to find the optimal solutions in terms of ϵ using the ϵ -constraint method and the Karush-Kuhn-Tucker (KKT) optimality conditions.

$$\begin{aligned} & \text{Minimize} \quad f_1(x) = x_1^2 - 4x_2 \\ & \text{Minimize} \quad f_2(x) = x_2^3 + x_1^2 \end{aligned}$$

Solution

Choosing minimize f_1 and constraining f_2 , the ϵ -constraint formulation is:

$$\quad \quad \quad (47)$$

$$\quad \quad \quad (48)$$

The Lagrangian is:

$$\text{[Redacted]} \quad (49)$$

The KKT stationarity conditions $\nabla_{\mathbf{x}} \mathcal{L} = \mathbf{0}$:

$$\text{[Redacted]} \quad (50)$$

$$\text{[Redacted]} \quad (51)$$

The KKT conditions also require:

$$\mu \geq 0, \quad g(\mathbf{x}) \leq 0, \quad \mu g(\mathbf{x}) = 0 \quad (52)$$

From (50), either $x_1 = 0$ or $\mu = -1$.

Case A: $\mu = -1$

From (51):

$$\text{[Redacted]}$$

This has no real solution, so Case A is rejected.

Case B: $x_1 = 0$

From (51): $\mu = \frac{4}{3x_2^2}$, which requires $x_2 \neq 0$ and gives $\mu > 0$ automatically for all real $x_2 \neq 0$.

Since $\mu > 0$, by complementary slackness (52) the constraint must be active:

$$\text{[Redacted]} \quad (53)$$

Then from (51):

$$\text{[Redacted]} \quad (54)$$

For $\mu > 0$ we require $\varepsilon > 0$.

Combining $x_1 = 0$, (53), and (54):

$$\boxed{x_1^* = 0, \quad x_2^* = \varepsilon^{1/3}, \quad \lambda^* = \frac{4}{3\varepsilon^{2/3}}} \quad (55)$$

The corresponding optimal objective values are:

$$\boxed{f_1^* = x_1^{*2} - 4x_2^* = 0 - 4\varepsilon^{1/3} = -4\varepsilon^{1/3}} \quad (56)$$

$$\boxed{f_2^* = x_2^{*3} + x_1^{*2} = (\varepsilon^{1/3})^3 + 0 = \varepsilon} \quad (57)$$

Note that (57) confirms the active constraint $f_2^* = \varepsilon$ as expected. The Pareto-optimal front is parametrized by ε via (55).

Since $f_2^* = \varepsilon$ and $f_1^* = -4\varepsilon^{1/3}$, we get $\varepsilon^{1/3} = -f_1^*/4$, so $\varepsilon = (-f_1^*/4)^3$, giving:

$$\boxed{f_2 = -\frac{f_1^3}{64}} \quad (58)$$

which gives the Pareto front relation between f_1 and f_2 .

Q5: It is desired to solve the with Python.

$$\begin{aligned} \min \quad & f_1(\mathbf{x}) = 1 - \exp \left(- \sum_{i=1}^n \left(x_i - \frac{1}{\sqrt{n}} \right)^2 \right) \\ \min \quad & f_2(\mathbf{x}) = 1 - \exp \left(- \sum_{i=1}^n \left(x_i + \frac{1}{\sqrt{n}} \right)^2 \right) \end{aligned}$$

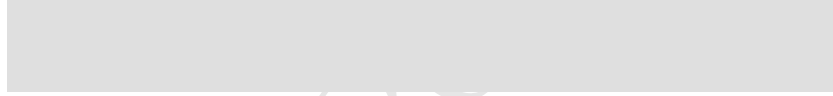
Subject to

$$-4 \leq x_i \leq 4; \quad i = 1, 2, 3$$

Solution

Both objectives are bounded in $[0, 1)$. The two objectives conflict with each other: f_1 is minimized when every x_i approaches $+1/\sqrt{n}$, while f_2 is minimized when every x_i approaches $-1/\sqrt{n}$. No single point can minimize both simultaneously, which gives rise to a non-trivial Pareto front.

The analytical Pareto-optimal set of this problem is the line segment



and the corresponding Pareto front is a continuous, non-convex curve in the (f_1, f_2) plane.

The NSGA-II algorithm was implemented using the `pymoo` Python library. The parameter settings used in this study are listed in Table 1.

Table 1: NSGA-II parameters used in this study.

Parameter	Value
Library	<code>pymoo</code> 0.6.1.6
Number of decision variables	3
Variable bounds	$[-4, 4]$ for each x_i
Population size	100
Number of generations	200
Crossover operator	SBX, $p_c = 0.9$, $\eta_c = 15$
Mutation operator	Polynomial (PM), $\eta_m = 20$
Mutation probability	$1/n$ (one gene per individual)
Duplicate elimination	Enabled
Random seed	1

Pareto Front in Objective Space

Figure 1 compares the 100 non-dominated solutions returned by NSGA-II with the analytical Pareto front. The approximation lies almost exactly on the true non-convex curve and is distributed uniformly across the full extent from $(f_1, f_2) = (0, 0.98)$ to $(0.98, 0)$.

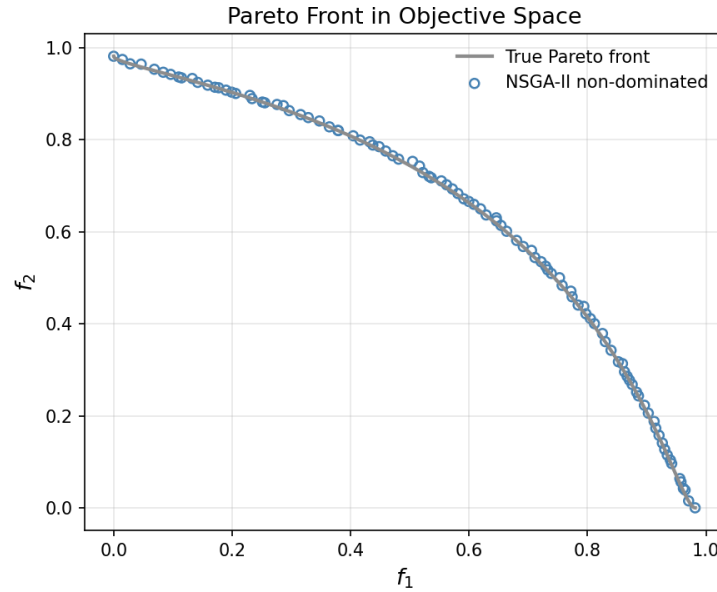


Figure 1: NSGA-II non-dominated set versus the analytical Pareto front in the objective space (f_1, f_2) .

Pareto-Optimal Set in Decision Space

Figure 2 projects the obtained solutions onto the x_1 - x_2 plane. Solutions cluster tightly around the diagonal $x_1 = x_2 = x_3$, confirming convergence to the correct region of decision space. The spread covers the full interval $[-0.577, 0.577]$.

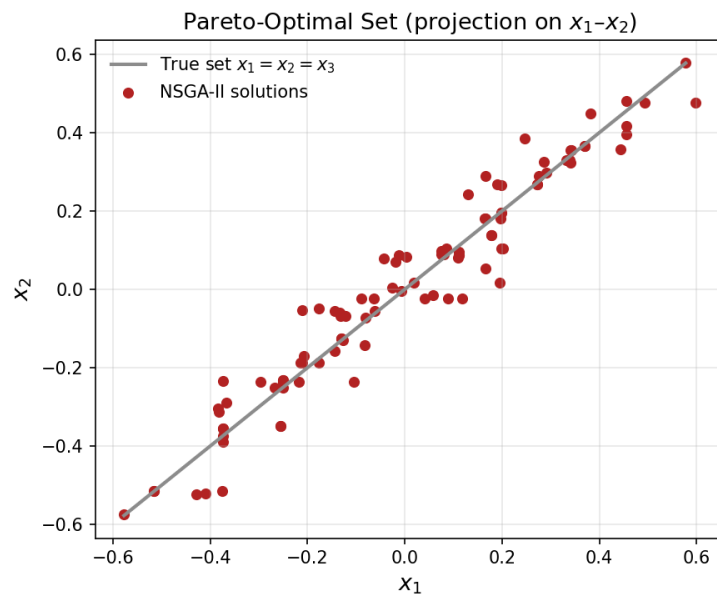


Figure 2: Pareto-optimal set projected onto the x_1 - x_2 plane. The grey line is the theoretical set $x_1 = x_2 = x_3$.

Convergence History

Figure 3 reports the hypervolume and IGD at every generation. Both curves stabilize around generation 25, showing that the population reaches a high-quality approximation of the Pareto front early in the run.

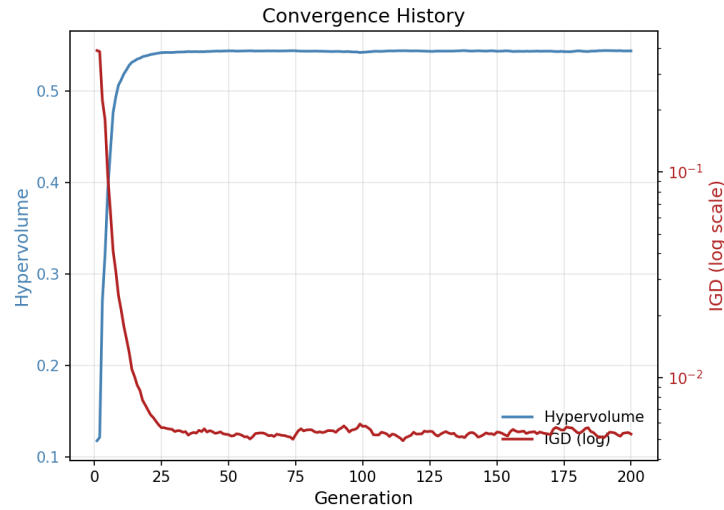


Figure 3: Hypervolume (left axis, blue) and IGD (right axis, red, log scale) over 200 generations.

Final Performance Metrics

Table 2 reports the indicator values at the final generation.

Table 2: Performance indicators at generation 200.

Indicator	Value
Hypervolume, ref. point (1.1, 1.1)	0.5441
Inverted Generational Distance (IGD)	0.005292
Generational Distance (GD)	0.002567
Spacing (Schott)	0.005876
Non-dominated solutions in final population	100

Convergence at Selected Generations

Table 3 tracks how the indicators and the size of the non-dominated set evolve at key generations.

Table 3: Indicator values at selected generations.

Gen	HV	IGD	GD	# non-dom.
1	0.1180	3.89×10^{-1}	3.13×10^{-1}	3
5	0.3910	9.62×10^{-2}	7.22×10^{-2}	13
10	0.5124	2.13×10^{-2}	1.43×10^{-2}	48
25	0.5422	5.69×10^{-3}	3.34×10^{-3}	100
50	0.5442	5.26×10^{-3}	2.43×10^{-3}	100
100	0.5427	5.77×10^{-3}	3.18×10^{-3}	100
150	0.5437	5.41×10^{-3}	2.55×10^{-3}	100
200	0.5441	5.29×10^{-3}	2.57×10^{-3}	100

Representative Non-Dominated Solutions

Table 4 lists 12 evenly-spaced solutions from the final non-dominated set (sorted by f_1). The boundary solutions recover $x_i = \pm 0.5774$, matching the theoretical endpoints of the Pareto set.

Table 4: Twelve representative non-dominated solutions, sorted by f_1 .

x_1	x_2	x_3	f_1	f_2
0.5773	0.5781	0.5773	0.0000	0.9817
0.3810	0.4479	0.2823	0.1327	0.9334
0.2755	0.2890	0.2744	0.2336	0.8896
0.1959	0.1804	0.1903	0.3642	0.8282
0.1116	0.0879	0.1525	0.4711	0.7654
0.0797	0.0891	-0.0242	0.5717	0.6933
0.0402	-0.0225	-0.0648	0.6538	0.6140
-0.1330	-0.0681	-0.0718	0.7388	0.5095
-0.2115	-0.1880	-0.1020	0.8117	0.4004
-0.2505	-0.2324	-0.2835	0.8753	0.2681
-0.3738	-0.3546	-0.3657	0.9302	0.1270
-0.5768	-0.5747	-0.5782	0.9816	0.0000